

WEEK 1

Task1

To Install Linux VM, Configure GCC, Setup Git Repository,
Create Project Structure, Understand Shell Architecture,
Build Initial Makefile

```
imvarls@LAPTOP-VQ4D70L7:~/OSSP$ tree
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c

9 directories, 18 files
imvarls@LAPTOP-VQ4D70L7:~/OSSP$ |
```

Task2

To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing

Source code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {

    pid_t pid;
    printf("Current Process PID : %d\n", getpid());
    printf("Current Process PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("\nChild will execute 'ls' using exec()\n");
        sleep(2);

        execl("/bin/ls", "ls", "-l", NULL);
        perror("exec failed");
        exit(1);
    } else {
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        printf("\nParent waiting for child...\n");
        wait(NULL);
        printf("\nChild process has terminated.\n");
        printf("Parent process completed.\n");
    }
    return 0;
}
```

```
Current Process PID : 525
Current Process PPID : 304
Parent PID : 525
Child PID : 526

Parent waiting for child...
Child PID : 526
Child PPID : 525
```

```
-rw-r--r--  1 root root    45 Aug  7 14:14 source.txt
-rwxr-xr-x  1 root root 16408 Jul 31 15:17 source_code
-rw-r--r--  1 root root   798 Jul 31 15:17 source_code.c
-rw-r--r--  1 root root   8102 Aug  7 14:28 trace.txt
```

```
Child process has terminated.
Parent process completed.
```

Task3

To Create Main Loop, Display Prompt, Read User Input,
Handle Exit Conditions, Design Control Flow Diagram, Test
Interactive Loop

```
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myshell> ");
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\nInput error or EOF detected.\n");
            break;
        }
        input[strcspn(input, "\n")] = '\0';
        if (strlen(input) == 0) {
            continue;
        }
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }
        if (strcmp(input, "hello") == 0) {
            printf("Hello! Welcome to the interactive shell.\n");
        }
        else if (strcmp(input, "help") == 0) {
            printf("Available commands:\n");
            printf("hello - Display greeting\n");
            printf("help  - Display help\n");
            printf("exit  - Exit the shell\n");
        }
        else {
            printf("Unknown command: %s\n", input);
        }
    }

    printf("Program terminated.\n");

    return 0;
}
```

```

myshell> hello
Hello! Welcome to the interactive shell.
myshell> help
Available commands:
hello - Display greeting
help  - Display help
exit  - Exit the shell
myshell> abc
Unknown command: abc
myshell> exit
Exiting shell...
Program terminated.

```

Task4

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```

#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define BUFFER_SIZE 100

int main() {
    struct termios oldt, newt;
    char buffer[BUFFER_SIZE];
    int index = 0;
    char ch;
    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);
    printf("Type a command and press Enter.\n");
    printf("Type 'exit' to quit.\n\n");

    printf("> ");

    while (1) {
        read(STDIN_FILENO, &ch, 1);
        if (ch == '\n' || ch == '\r') {
            buffer[index] = '\0';

            printf("\nYou entered: %s\n", buffer);
            if (strcmp(buffer, "exit") == 0) {
                break;
            }
            if (strcmp(buffer, "hello") == 0) {
                printf("Hello! Welcome.\n");
            }
            else if (strcmp(buffer, "help") == 0) {
                printf("Commands: hello, help, exit\n");
            }
            else if (index == 0) {
                printf("No input entered.\n");
            }
            else {

```

```

        printf("No input entered.\n");
    } else {
        printf("Command received successfully.\n");
    }
    index = 0;
    printf("> ");
}
else if (ch == 127 || ch == 8) {
    if (index > 0) {
        index--;
        printf("\b \b");
        fflush(stdout);
    }
}
else if (ch >= 32 && ch <= 126) {
    if (index < BUFFER_SIZE - 1) {
        buffer[index++] = ch;
        printf("%c", ch);
        fflush(stdout);
    }
}
}
tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
printf("\nProgram terminated.\n");

return 0;
}

```

OUTPUT

```

> help
You entered: help
Commands: hello, help, exit
> helpp
You entered: helpp
Command received successfully.
> exit
You entered: exit

Program terminated.
root@DESKTOP-1MNG33K:~#

```